



Decoding mutual fund performance: Dynamic return patterns via deep learning[☆]

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ABSTRACT

This paper applies the Temporal Fusion Transformer (TFT) model to learn dynamic time-series patterns in mutual fund performance and to assess whether these patterns predict future alpha. I summarize the model's cross-sectional ranking power using diagnostic portfolio spreads: a top-minus-bottom decile exhibits an annualized Carhart four-factor alpha spread of 2.8%, with dispersion persisting for up to four years. In panel regressions controlling for standard predictors and fund and time fixed effects, TFT forecasts improve explanatory power by more than 25% in adjusted R^2 . Leveraging TFT's interpretable outputs, I show that historical fund returns receive the largest weight (about 29%), their importance displays earnings-cycle seasonality, and attention to past observations rises by 46% during crisis periods. Using fund-by-month variable-importance weights, I define fund-specific informativeness states and construct conditional skill measures that predict and persist precisely when the same signal becomes informative again, beyond coarse macro conditioning. Together, these results provide an alternative explanation for why unconditional performance persistence appears weak: skill is episodic and becomes visible when a manager's key signals regain relevance.

"History doesn't repeat itself, but it often rhymes"

[—Mark Twain]

1. Introduction

History often repeats itself, though rarely in exactly the same way. For example, the 2020s share notable parallels with the "roaring" 1920s: both periods followed a global pandemic and witnessed rapid technological advancement, despite significant differences between the two eras.² In the mutual fund industry, experienced managers similarly learn from recurring market and economic episodes and adjust their strategies accordingly.³ This perspective offers an alternative interpretation of two well-known empirical facts: most funds do not generate

superior performance and past performance is not a dependable predictor of future performance (e.g., Fama and French, 2010; Barras et al., 2010). If funds deploy different, state-dependent strategies over time, then the mapping from past outcomes to future performance can be nonlinear and regime-specific, making persistence difficult to detect with traditional linear models.

This paper sets out to identify and learn these dynamic time-series patterns in fund returns, assess whether they help predict future performance (or managerial skill), and connect the patterns to the way funds respond to changing market conditions. While the classical mutual fund literature emphasizes cross-sectional factor models (Carhart, 1997; Fama and French, 2010) and recent machine learning work has made progress using fund characteristics and other cross-sectional signals (Kaniel et al., 2023; Li and Rossi, 2020), there remains scope to study

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² "Roaring"? Not so fast", Cristina Lindblad, February 1, 2021, Bloomberg Businessweek.

³ See, for example, *The Most Important Thing: Uncommon Sense for the Thoughtful Investor*, 2011, Howard Marks, Columbia Business School Publishing, "Déjà vu all over again", January 10, 2019, Andrew Pastor, EdgePoint Investment Group, and "ARK Invest's Wood expects market rotation back to growth stocks", September 14, 2021, David Randall, Reuters.

how performance evolves as strategies and information environments change over time.

To take this time-series perspective seriously, I employ a state-of-the-art sequence-to-sequence deep learning model, the Temporal Fusion Transformer (TFT). Unlike single-period forecasting, TFT predicts an entire sequence of future fund returns jointly, which is well suited to capturing multi-horizon dynamics and recurring patterns. The model also integrates dynamic (e.g., macroeconomic indicators), deterministic (e.g., fund age), and static (e.g., fund style) inputs through distinct pathways.⁴ Most importantly for economic interpretation, TFT produces exportable attention and variable-importance matrices that summarize (i) which past periods the model relies on for forecasting and (ii) which variables are most informative for a given fund at a given point in time. These intermediate outputs make it possible to move beyond “black-box” prediction and study when and why skill becomes economically relevant.

I first evaluate whether TFT meaningfully distinguishes funds based on predicted conditional performance. I summarize cross-sectional ranking ability using diagnostic portfolio spreads: a top-minus-bottom decile spread corresponds to an annualized Fama–French–Carhart four-factor alpha of 2.8%. Because shorting mutual funds is generally not feasible for typical investors, I interpret this spread as a diagnostic summary of conditional ranking power rather than a tradable strategy. The dispersion in conditional performance is persistent, remaining statistically and economically significant for up to four years after portfolio formation. Consistent with this ranking evidence, TFT forecasts also improve predictive-regression fit: after controlling for historical performance, fund characteristics, and other indicators (including fund and time fixed effects), the model increases the adjusted R-squared by more than 25%, indicating that it captures time-varying information about performance not explained by conventional predictors.

The interpretable outputs then clarify what the model learns and how that information maps into economic mechanisms. Variable importance assigns the largest weight to a fund’s own return history (nearly 29% of decision weight), followed by macro and market variables such as equity market returns, fixed-income spreads, inflation, and Treasury yields. The importance of historical alpha exhibits pronounced seasonality aligned with corporate earnings cycles. The attention mechanism places systematically higher weight on crisis observations: attention weights during financial crises are approximately 46% higher than in non-crisis periods. These patterns suggest that both the content of past returns and the information environment in which those returns are realized are central to understanding subsequent performance.

Finally, I use the fund-by-month, variable-specific importance weights to define informativeness states and to construct model-based conditional skill measures. These measures capture a fund’s abnormal performance during periods when a particular signal is highly informative according to the model, and they predict future performance specifically when those signals become informative again. Conditional variables span both macroeconomic indicators (market return, inflation, term spread, default spread) and fundamental proxies (fund past returns and month-of-year). The resulting conditional skill measures are persistent for up to four years and remain predictive even after controlling for direct macro conditioning, indicating that the persistence is not driven by macro conditions alone. Overall, the evidence supports a state-dependent view of skill: managers’ abilities surface when their preferred signals regain relevance, which helps reconcile the lack of unconditional performance persistence with the presence of conditional, time-varying skill.

This paper contributes to the emerging machine-learning literature on mutual funds and to the broader evidence on conditional fund skill.

Recent studies apply non-linear methods to fund characteristics and holdings to improve cross-sectional fund selection (e.g., Kaniel et al., 2023; Li and Rossi, 2020). DeMiguel et al. (2023) further show that characteristic-based machine-learning signals can translate into economically meaningful performance in implementable long-only portfolios after accounting for key frictions.

This paper is complementary but distinct in both object and mechanism. While Kaniel et al. (2023) and DeMiguel et al. (2023) primarily study cross-sectional fund selection using relatively static characteristics (and emphasize implementable long-only strategies), I focus on the temporal structure of fund performance and use a sequence-to-sequence model to learn recurring, state-dependent return patterns. This time-series emphasis allows me to summarize conditional ranking power using diagnostic portfolio spreads and, crucially, to use attention and variable-importance outputs to define fund-specific informativeness states. The resulting conditional skill measures predict future performance precisely when a fund’s preferred signals regain informativeness, remain persistent for up to four years, and are not explained by coarse macro conditioning alone.

Second, this paper contributes to the literature on time-varying fund skill and conditional performance predictability.⁵ Prior work shows that fund performance and risk exposures can depend on conditioning information and evolve with the economic environment. I complement these approaches by using model-implied informativeness states to discipline when conditioning should matter: rather than pre-specifying a small set of instruments or regimes, I let the data identify which signals are informative for which funds and when.

The key contribution is a set of fund-by-signal conditional skill measures built directly from the TFT variable-importance outputs. These measures isolate abnormal performance during periods when a particular variable is judged informative for a given fund and predict future performance specifically when that same signal becomes informative again. This framework yields a more granular, manager-specific view of conditional skill than aggregate macro conditioning and links conditional predictability to an explicit, interpretable mechanism.

Third, this paper contributes to the debate on mutual fund performance persistence. The classical literature (e.g., Jensen, 1968, Elton et al., 1993, Carhart, 1997, Busse et al., 2010, and Fama and French, 2010) finds limited evidence that past performance predicts future performance in an unconditional sense.

I provide a mechanism-based reconciliation. If funds deploy strategies whose effectiveness depends on which signals are currently informative, then unconditional persistence can be weak even when managers possess skill that is real but episodic. Using the TFT to learn time-series return patterns and to identify fund-specific informativeness states, I show that persistence concentrates in the forecasted conditional component: conditional skill measures predict future abnormal performance when the same signals regain informativeness, and this predictability persists for up to four years while remaining distinct from coarse macro regimes.

2. Data, variable construction, and sample overview

2.1. Data sources

This study draws data from several complementary sources. Mutual fund returns, total net assets (TNA), expense ratios, turnover ratios, investment objectives, and other fund characteristics come from the CRSP Survivorship Bias-Free Mutual Fund database. Portfolio holdings data are sourced from the Thomson Reuters Mutual Fund Holdings (s12) database. I link these two databases using the MFLINKS tables

⁴ Dynamic variables are time-varying and subject to random variations each period. Deterministic variables follow a predetermined path. Static variables remain constant over time.

⁵ See, for example, Ferson and Schadt (1996), Moskowitz (2000), Cooper et al. (2004), Mamaysky et al. (2008), Glode and Green (2011), and Kacperczyk et al. (2014).

provided by Wharton Research Data Services (WRDS). All macroeconomic variables are obtained from Federal Reserve Economic Data (FRED).

The analysis focuses on active U.S. equity funds spanning January 1990 through December 2019.⁶ Following the established criteria in Kacperczyk et al. (2008), I identify domestic equity funds while excluding various specialized fund categories.⁷ Specifically, I remove ETFs, fixed income, international, money market, sector, index, target-date, and balanced funds from the sample.⁸ To address potential omission bias (Elton et al., 2001) and incubation bias (Evans, 2010), I exclude observations prior to funds' first offer dates, those with missing fund names in the CRSP database, and funds with TNA below \$5 million. The final sample comprises 3717 unique funds with 500,113 fund-month observations.

2.2. Variable construction

2.2.1. Fund performance and characteristics

To measure fund performance, I compute alphas based on rolling window estimates of factor betas. For each fund-month observation, I estimate betas using the previous 24 months of returns against the CRSP value-weighted excess market return (Mktrf), size (SMB), book-to-market (HML), and momentum (UMD) factors from Ken French's website. I then use these estimated betas to risk-adjust the fund's current month excess return.

Since the CRSP Mutual Fund database reports multiple share classes as separate entities, I aggregate them to the fund level. Total net assets (TNA) represents the sum of assets across all share classes of a fund (in \$mm). For variables such as cash holdings, turnover ratio, expense ratio, and management fees, I calculate TNA-weighted averages across all share classes, with values expressed in percentage points. Manager tenure reflects the number of years since a portfolio manager was hired; when multiple managers oversee a fund, I use the longest tenure. I create a load dummy variable that equals one if any share class of the fund charges a load fee, and zero otherwise.

To identify fund managers' unobservable skill, I follow the extant literature and employ the return gap measure (Kacperczyk et al., 2008). This monthly measure represents the difference between a fund's realized gross return and the hypothetical return on its most recently disclosed portfolio holdings. I define Return Gap as the average monthly return gap over the previous 12 months.

2.2.2. Macroeconomic and market variables

To capture the macroeconomic environment that may influence fund performance, I incorporate a comprehensive set of macroeconomic variables that prior literature has identified as significant predictors of security returns and risks. These include: (1) Industrial Production Index, (2) Consumer Price Index, (3) Crude Oil Price (WTI), (4) 3-month Treasury bill rate, (5) Term spread between 10-year Treasury and 3-month Treasury bill yields, (6) Default spread between Baa and Aaa corporate bond yields, and (7) NBER recession indicators. For Industrial Production and Consumer Price indices, I measure the percentage change from a year ago, while the remaining variables (except NBER recession indicators) are calculated as percentage changes from the previous month.

⁶ The sample begins in 1990 because several key macroeconomic variables, including VIX and Oil Price data, become consistently available after this date.

⁷ For detailed selection criteria, see Kacperczyk et al. (2008), Appendix A, page 2412.

⁸ Following Jones and Mo (2021), I identify index funds through both the CRSP index fund flag and by searching fund names for keywords such as *Exchange-traded*, *Etf*, *Dfa*, *Index*, *Inde*, *Indx*, *Inx*, *Idx*, *Dow Jones*, *Ishare*, *S&P 500*, *Wilshire*, *Russell*, *Russ*, and *MSCI*. Target-date funds are excluded by searching for terms like *2055*, *2050*, *2045*, *2040*, *2035*, *2030*, *2025*, *2020*, *2015*, *2010*, *2005*, and *Target*.

Table 1
Summary statistics.

Variables	Mean	Median	Std	P25	P75
Fund return & characteristics					
<i>Alpha</i>	−0.12	−0.11	−0.11	−0.97	0.74
<i>Flow</i>	1.75	−0.20	−0.20	−1.34	1.49
<i>TNA</i>	1185.01	195.90	195.90	47.30	815.90
<i>Load</i>	0.56	1.00	1.00	0.00	1.00
<i>Cash</i>	5.00	2.42	2.42	0.70	5.49
<i>Expense</i>	1.20	1.15	1.15	0.90	1.46
<i>Management fee</i>	0.58	0.72	0.72	0.50	0.88
<i>Turnover</i>	0.92	0.62	0.62	0.33	1.07
<i>Manager Turnover</i>	0.60	0.67	0.43	0.25	1.00
<i>Total number of funds</i>	3717				
Macro variables					
<i>Industry production</i>	1.90	2.66	2.66	0.71	4.03
<i>Inflation</i>	2.45	2.49	2.49	1.70	3.07
<i>Oil price</i>	8.73	5.20	5.20	−12.28	26.67
<i>T-Bill yield</i>	3.74	0.00	0.00	−3.33	3.85
<i>Term spread</i>	0.01	−2.00	−2.00	−15.00	14.00
<i>Default spread</i>	−0.04	−1.00	−1.00	−4.00	4.00
<i>Crisis Dummy</i>	0.09	0.00	0.00	0.00	0.00
<i>VIX</i>	1.10	−1.38	−1.38	−8.89	6.79
<i>Value-weighted return</i>	0.87	1.34	1.34	−1.69	3.53
<i>S&P 500 return</i>	0.71	1.11	1.11	−1.74	3.25
<i>Market risk premium</i>	0.67	1.18	1.18	−1.90	3.37
<i>Enhanced Sentiment</i>	0.03	0.08	2.07	−1.37	1.46

This table presents summary statistics for the sample of active US domestic equity mutual funds from 1990 to 2019. Fund returns and characteristics are reported at the fund level, while macro-level variables are calculated monthly based on information available in the previous month. All variables are defined in Appendix A.

Additionally, I incorporate market variables that potentially influence fund managers' investment decisions. These market indicators include the percentage change in the CBOE volatility index (VIX), returns on the CRSP total-return value-weighted index, and S&P 500 index returns. I also include returns from prominent factor models: the size (SMB), value (HML), profitability (RMW), and investment (CMA) factors from the Fama and French five-factor model, as well as the size (R_{ME}), investment (R_{IA}), profitability (R_{ROE}), and expected growth (R_{EG}) factors from the Hou et al. (2015) q-factor model.

Complete definitions and sources for all variables used in the machine learning models are provided in Appendix A, with summary statistics presented in Table 1. All variables are constructed on a monthly basis using information available at the previous month-end, and any potentially unbounded variables are winsorized at the 1% extremes to mitigate the influence of outliers.

3. Methodology

3.1. Predicting mutual fund performance

Traditional approaches to predicting mutual fund performance have primarily relied on single-period predictive regression models. These models typically incorporate theoretically motivated variables such as return gap (Kacperczyk et al., 2008), active share (Cremers and Petajisto, 2009), and risk shifting (Huang et al., 2011) to forecast future fund performance. However, such models often fail to utilize the entire historical time-series of fund returns, largely due to the limitations of linear regression when handling numerous potentially correlated variables. While conventional time-series models like ARIMA (Box et al., 1974) and exponential smoothing (Hyndman et al., 2008) can capture temporal patterns such as trends and seasonality, they typically impose rigid structural assumptions and perform optimally only when the underlying time-series structure is well understood. These limitations create an opportunity for more flexible approaches that can better capture the complex dynamics of mutual fund performance

without requiring strong a priori assumptions about the data-generating process.

Deep neural networks (DNNs) have demonstrated superior performance in time-series forecasting by extracting complex patterns without requiring structural assumptions about the data (Rangapuram et al., 2018, Salinas et al., 2020, Wen et al., 2017). However, standard DNN architectures face significant challenges when applied to financial data. Financial datasets typically feature small sample sizes with weak signal-to-noise ratios (Israel et al., 2020), making them susceptible to distortion from irrelevant inputs. Moreover, conventional architectures often fail to properly account for the heterogeneity of input types, simply concatenating static features with time-dependent variables. The “black-box” nature of these models further complicates interpretation, as predictions result from complex nonlinear interactions among numerous parameters. Addressing these limitations requires specialized architectures designed specifically to handle financial data’s unique characteristics while providing interpretable results.

3.2. Temporal fusion transformers model

In this paper, I implement the Temporal Fusion Transformer (TFT) model developed by Lim et al. (2021), leveraging its robust architecture to enhance the analysis of mutual fund performance patterns (Fig. 1). TFT distinguishes itself through its ability to simultaneously produce multi-horizon forecasts, which allows a comprehensive visualization of fund performance trajectories over an extended time frame. This capability is particularly advantageous for financial applications, where insights into evolving performance patterns substantially benefit strategic decision-making.

The architecture of TFT comprises four principal elements, each explicitly designed to optimize predictive accuracy and model interpretability in financial time series contexts:

1. Variable Selection Module: TFT employs a dynamic variable selection network that assigns interpretable, period-specific weights to each input variable via vector multiplications. This ensures that the model selectively emphasizes only the most relevant variables at each time step, effectively filtering noise and directly indicating variable importance.

2. LSTM Encoder and Decoder: Leveraging the strengths of Long Short-Term Memory (LSTM) networks, TFT encodes historical observations and static covariates into standardized vectors. These vectors capture both temporal dependencies and cross-variable correlations, subsequently serving as robust inputs for further predictive analysis.

3. Attention Matrix: TFT integrates a self-attention mechanism, generating an attention matrix that explicitly highlights influential historical time periods for forecasting specific future horizons. This mechanism enhances the interpretability of temporal dynamics, providing direct visibility into which historical points exert significant predictive influence.

4. Quantile Loss Optimization: TFT uses a quantile loss function to generate accurate prediction intervals. The quantile loss ensures that predictions capture a range of possible outcomes, which is critical for effective risk management and informed financial decision-making.

Collectively, these design elements enable TFT not only to achieve superior forecasting accuracy but also to provide transparent, actionable insights into the variables and temporal structures driving mutual fund performance. For a comprehensive mathematical formulation and detailed architectural description of the Temporal Fusion Transformer (TFT) model, including the specific equations governing each component, please refer to Appendix B.

3.3. Model evaluation

To assess the predictive performance of fund alpha forecasts, I employ an out-of-sample R-squared methodology following Gu et al.

(2020). Specifically, I calculate the out-of-sample R^2 as:

$$R^2_{OOS} = 1 - \frac{\sum_{(i,t) \in \tau_{OOS}} (r_{i,t+1} - \hat{r}_{i,t+1})^2}{\sum_{(i,t) \in \tau_{OOS}} r_{i,t+1}^2}, \quad (1)$$

where i and t denote the fund and month indices, respectively, and τ_{OOS} represents the out-of-sample period that was never used in model estimation or hyperparameter tuning. The R^2_{OOS} metric pools prediction errors across funds and over time to provide a comprehensive panel-level assessment of each model. Since the denominator uses the sum of squared excess returns without demeaning, this measure quantifies the proportional reduction in mean squared forecast error (MSFE) achieved by the model relative to a naive benchmark that simply forecasts zero.

3.4. Interpretable variable importance and attention

A key advantage of the TFT model is its ability to provide interpretable variable selection for each data type, including dynamic inputs, deterministic inputs, and static inputs. The model generates several interpretable outputs that will be crucial for understanding the results in subsequent sections.

For historical inputs, the variable selection weights are calculated as:

$$\mathbf{w}_{i,j,t} = \text{Softmax}(f(\epsilon_1, \dots, \epsilon_j, \mathbf{c}_s)), \quad (2)$$

where $\mathbf{w}_{i,j,t}$ represents the vector of variable selection weights for each historical variable j (from the set $\{\epsilon_1, \epsilon_2, \dots, \epsilon_k\}$) for fund i at time t , $\mathbf{c}_s$ denotes the vector of all static inputs, and f is a gating module that integrates multiple variables.⁹ The softmax function normalizes these inputs into a probability vector where components sum to one, effectively quantifying the relative importance of each variable.

These interpretable weights are then used to aggregate inputs into the subsequent layer:

$$V_{i,t} = \sum_{j=1}^k \epsilon_{i,j,t} w_{i,j,t} \quad (3)$$

In addition to providing interpretable variable importance, the TFT model employs a self-attention mechanism to capture both short-term and long-term temporal relationships. This attention mechanism is formulated as:

$$A_{i,t} = g(V_{i,t}), \quad (4)$$

where g represents a function that encodes the aggregated features $V_{i,t}$ into a sequence-to-sequence layer followed by attention architectures. The resulting attention weights can be extracted to identify which historical periods are assigned greater importance in the prediction process, providing valuable insights into the model’s temporal focus.

4. Predicting mutual fund performance using machine learning models

4.1. Machine learning framework

This section presents the machine learning framework employed to predict sequences of mutual fund alphas across multiple future periods. The prediction model is defined as:

$$\alpha_{i,t} = \mathbf{h}(I_{t-1}) + \epsilon_{i,t}, \quad (5)$$

where $\alpha_{i,t} = (\alpha_{i,t+1}, \dots, \alpha_{i,t+T})$ represents the vector of fund alphas for the future T periods after period t , I_{t-1} denotes the public information set available at $t - 1$, and $\mathbf{h}(I_{t-1}) = (h_{i,t+1}(I_{t-1}), \dots, h_{i,t+T}(I_{t-1}))$ is a vector-valued function that approximates the expected future fund alphas. The prediction horizon T specifies the length of the sequence

⁹ For static variables, the context vector $\mathbf{c}_s$ is omitted from the calculation.

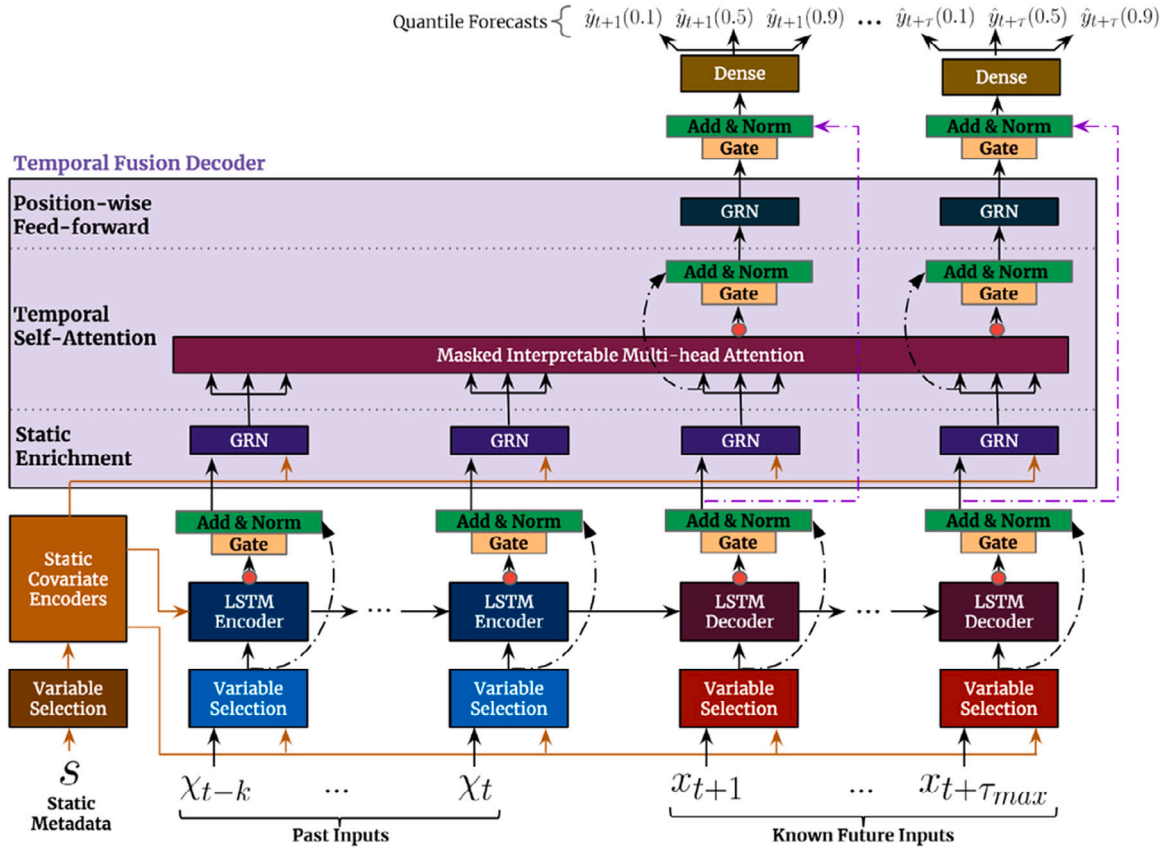


Fig. 1. Basic structure of Temporal Fusion Transformer (TFT) model.

This figure illustrates the architecture of the Temporal Fusion Transformer (TFT), which combines interpretable attention, gating, and variable selection across static and time-varying inputs to support multi-horizon forecasting. Reproduced from Lim et al. (2021).

to be forecasted. Throughout the paper, these fund alphas represent the “target variable” in the estimation process.

The information set I_{t-1} comprises historical values of various predictors $\{z_{i,s} : t - T^* \leq s < t\}$, where the estimation horizon T^* indicates the maximum historical time span considered. These predictors fall into three distinct categories: (1) dynamic inputs that covary with the target variables over time (e.g., macroeconomic variables), (2) static inputs whose values remain constant over time (e.g., fund style), and (3) deterministic inputs representing characteristics that vary with time but have values known in advance (e.g., fund age).

The predictors encompass a comprehensive set of variables potentially influencing fund performance. For dynamic inputs, I include fund characteristics (flow, TNA, cash holdings, and equity holdings), macroeconomic variables (industrial production, inflation, oil price, risk-free rate, default spread, and term spread), and market-related variables (S&P 500 return, VIX, value-weighted index return, NBER crisis dummy, factor returns from the Fama–French five-factor model and Hou et al. (2015) q-factor model, and momentum factor return). Static inputs comprise fund style, load, and management fee, while deterministic inputs include the upcoming month of the year and manager tenure. To enhance model efficiency and prediction accuracy, all unbounded inputs and target variables are standardized by month (zero mean and unit standard deviation) before implementation. Importantly, due to their distinct characteristics, the TFT model processes dynamic, static, and deterministic variables through separate pathways.

4.2. Data partitioning, tuning, and sensitivity analysis

To ensure robust model evaluation while preventing look-ahead bias, I implement a rigorous validation framework that addresses

two critical challenges in time-series forecasting: avoiding regime-specific calibration bias and preventing information leakage across funds. The framework combines rolling-window cross-validation with cross-sectional fund splitting in a complementary manner.

I employ a rolling-window cross-validation scheme with k folds that span the entire sample period.¹⁰ Each fold shifts the training window forward by 12 months while maintaining the same window length. For instance, the first fold trains on 1990–2009 data, the second fold trains on 1991–2010 data, and this pattern continues sequentially. This rolling structure serves two purposes: it allows the model to adapt to evolving market dynamics over time, and it generates multiple out-of-sample predictions for overlapping time periods, which can be averaged to create more stable forecasts.

Within each fold’s training period, I further partition the data by randomly assigning 80% of funds to a training subset and 20% to a validation subset, using a fixed random seed for reproducibility. Critically, this split is based on fund identifiers, not time periods. This means the training subset uses 80% of funds over the fold’s entire training window (e.g., all months from 1990–2009 for those selected funds in Fold 1), while the validation subset uses the remaining 20% of funds over the same time window. This cross-sectional approach provides three key advantages. First, it prevents fund-level information leakage—the model never sees data from validation funds during training, testing its ability to generalize to completely unseen funds. Second, both training and validation subsets contain observations from

¹⁰ Given the 1990–2019 sample period and the requirement for sufficient training data, I implement $k = 10$ folds, with each fold using a 240-month (20-year) training window.

Table 2
Performance comparison of machine learning models.

TFT	OLS	Ridge	Decision tree	AdaBoost	Random forest	NN2	NN3
0.3584	0.0272	0.0052	0.0039	0.0486	0.0363	0.0749	0.0392

This table reports the out-of-sample R^2_{OOS} based on out-of-sample predictions of different models: (i) TFT model, (ii) OLS model, (iii) ridge OLS regression model, (iv) decision tree model, (v) AdaBoost model, (vi) random forest model, (vii) neural network with two hidden layers (32 and 16 neurons), and (viii) neural network with three hidden layers (32, 16, and 8 neurons).

the same calendar periods, including normal markets, crises, and recoveries, which prevents hyperparameter optimization from overfitting to any specific market regime (e.g., calibrating solely on crisis periods). Third, each fund's time-series structure remains intact, preserving the sequential dependencies crucial for forecasting.

The framework maintains strict information controls at two levels. Within each fold, training and validation funds are completely disjoint sets—no fund appears in both. This ensures the model learns generalizable patterns common across fund managers rather than memorizing idiosyncrasies specific to individual funds. Across all folds, chronological precedence is preserved: standardization parameters (i.e., means and standard deviations for each variable) are calculated only from each fold's training data and applied consistently to that fold's validation and test data. This ensures the model never uses future information in either parameter estimation or feature scaling.

For hyperparameter optimization, I employ a systematic random search across a predefined parameter space exploring diverse combinations of dropout rates, hidden layer sizes, batch sizes, learning rates, gradient clipping thresholds, and attention mechanisms. For each candidate configuration, I train the model on all k folds and compute the average validation loss across folds. The configuration achieving the lowest average validation loss is selected, making the hyperparameter selection more robust to idiosyncrasies of any single time period.

After selecting optimal hyperparameters, I train the final model on each fold using its full training period and generate predictions for the subsequent test period. Because the test periods across folds overlap (e.g., 2010–2019 appears in multiple folds' test periods), I create ensemble forecasts by averaging predictions from all folds covering each fund-month observation. This ensemble approach substantially improves prediction stability and out-of-sample performance.

To enhance model stability, I set the estimation horizon to 72 months and the prediction horizon to 12 months, requiring funds to have at least 84 months of continuous observations.¹¹ Untabulated sensitivity checks across plausible hyperparameter ranges (attention heads, sequence lengths, learning rates) produce virtually identical alphas (± 10 bps) and out-of-sample R^2 ($\pm 1\%$), confirming that the model learns generalizable patterns rather than regime-specific artifacts. Complete implementation details, including the specific fold configurations, hyperparameter ranges, and optimization protocols, are provided in [Appendix C](#).

4.3. Performance comparison

To evaluate the TFT model's performance, I conduct a comprehensive comparison against multiple modeling approaches: generalized linear models (OLS, Lasso, Ridge, and Elastic Net), tree-based models (Decision Tree, Ada Boost, and Random Forest), and feed-forward neural networks with varying complexity (two and three hidden layers). Unlike the TFT model, these alternative approaches don't differentiate between dynamic, static, and deterministic variables. Therefore, I concatenate all covariates across the full 72-month estimation horizon as inputs, resulting in a total of $72 \times (1 + 33) = 2448$ features. Additionally, since these alternative models lack multi-horizon forecasting

capabilities, I use the average of future 12-month returns as the target variable.

Table 2 presents the out-of-sample R^2 comparison across these methods, arranged from simplest to most sophisticated. Unsurprisingly, the linear OLS model achieves minimal predictive performance with an R^2 of just 0.02%, largely due to its inability to capture nonlinear relationships and complex intertemporal patterns. Generalized linear models like Ridge regression, despite their feature selection capabilities, show negligible improvement over OLS, suggesting highly nonlinear relationships among the predictors.

While decision trees are designed to model nonlinear interactions, a single tree yields an even lower R^2 of 0.004%, likely due to high variance leading to poor generalization. Ensemble methods like Ada Boost and Random Forest aggregate information from multiple weak learners, improving performance with R^2 values of 0.05% and 0.04%, respectively. Neural networks capture more complex predictor interactions, further improving the R^2 to 0.07% with a two-layer architecture. The TFT model, leveraging its specialized architecture discussed earlier, dramatically outperforms all alternatives with an R^2 of 0.35%—nearly five times better than the next best approach.

To further assess the TFT model's predictive power, I compare it against traditional mutual fund alpha predictors through panel regression analysis. I define Predicted Alpha as the TFT model's alpha forecast for a given fund-month combination and estimate the following panel regression with both year and fund fixed effects:

$$Alpha_{i,t+1} = \beta Alpha Predictor_{i,t} + \gamma Control_{i,year} + \alpha_i + \alpha_t + \epsilon_{i,t} \quad (6)$$

where control variables include *log(TNA)*, *Fund Flow*, *Cash Holdings*, *Expense Ratio*, *Management Fee*, and *Turnover Ratio*.

Table 3 presents these results. The coefficient of *Predicted Alpha* remains statistically significant at the 1% level across all specifications, even after controlling for established predictors like historical alpha and return gap. This suggests that the TFT model captures information orthogonal to traditional measures. To quantify this contribution, I calculate the proportion of adjusted R^2 attributable to Predicted Alpha by measuring the increase in adjusted R^2 when adding this variable to the regression model. Remarkably, Predicted Alpha consistently contributes approximately 20% of the model's overall explanatory power, even in the most comprehensive specification that includes all control variables and fixed effects.

To further validate that the model captures fund-level skill rather than just manager-specific traits, I examine the impact of manager turnover. As shown in the Online Appendix (Table OA.1 and Table OA.2), the model's predictive power and the conditional persistence of performance remain robust even after controlling for manager turnover. It suggests that the "skill" is likely capturing a fund-level strategy or institutional capability that persists even when individual managers change.

While **Table 3** establishes that Predictive Alpha contains information about future performance that is incremental to traditional predictors, the next step is to understand what the model is learning and when that information is economically meaningful. To do so, I use the model's variable-importance outputs to construct fund-by-month informativeness states—periods in which a given macroeconomic or fundamental variable is particularly informative for a specific fund. I then define V.I. Conditional Alpha as the fund's realized alpha during past high-informativeness periods and test whether performance persistence is concentrated precisely when the model identifies similar informative

¹¹ The results remain robust under different estimation horizons, including 60 months and 48 months.

Table 3
Regression analysis of model-predicted fund performance.

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variables	Alpha					
Predicted alpha	0.387*** (11.95)	0.399*** (13.99)	0.332*** (10.40)	0.344*** (10.57)	0.342*** (11.18)	0.355*** (11.46)
Historical alpha				−0.014*** (−3.34)		−0.014*** (−3.63)
Return gap					−3.061 (−0.77)	−3.100 (−0.77)
Log (TNA)	0.001 (0.27)	−0.123*** (−8.17)	−0.132*** (−8.66)	−0.133*** (−8.69)	−0.139*** (−9.45)	−0.140*** (−9.43)
Fund flow	0.400* (1.92)	−1.250*** (−4.64)	−1.284*** (−4.77)	−1.265*** (−4.72)	−1.340*** (−5.19)	−1.320*** (−5.12)
Cash holdings	−0.001 (−1.27)	−0.001 (−0.40)	−0.001 (−0.35)	−0.001 (−0.38)	−0.001 (−0.56)	−0.001 (−0.58)
Expense ratio	−7.025*** (−3.76)	−3.840 (−0.36)	1.801 (0.17)	1.957 (0.18)	−4.349 (−0.43)	−4.123 (−0.40)
Management fee	0.036* (1.66)	−0.067 (−0.99)	−0.054 (−0.84)	−0.056 (−0.86)	0.001 (0.02)	−0.000 (−0.01)
Turnover ratio	−0.018** (−2.51)	−0.011 (−1.59)	−0.009 (−1.33)	−0.009 (−1.33)	−0.010 (−0.54)	−0.009 (−0.53)
Fund fixed effect	No	Yes	Yes	Yes	Yes	Yes
Year fixed effect	Yes	No	Yes	Yes	Yes	Yes
Observations	106,067	106,067	106,067	106,067	100,787	100,787
Adjusted R-squared	0.007	0.009	0.011	0.011	0.011	0.011
Predicted Alpha						
R ² Contribution	84.35%	37.93%	23.95%	24.55%	24.71%	25.15%

This table reports the regression of future fund alpha on the predicted alpha from the TFT model, historical alpha over the previous 12 months, return gap, and other fund characteristics. Variables are defined in Appendix A. The *t*-statistics, in parentheses, are based on standard errors clustered by funds. ***, **, * denote statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

conditions going forward. Historical Alpha serves as an unconditional benchmark, and Macro Conditional Alpha provides a robustness check that distinguishes model-identified informativeness states from coarse, economy-wide macro regimes.

4.4. Diagnostic portfolio evidence of performance dispersion

To summarize the model's ability to rank mutual funds by predicted conditional performance, I construct diagnostic portfolio sorts based on predicted alphas. These portfolios are not intended to represent feasible investment strategies, but rather to provide a transparent and familiar summary of cross-sectional dispersion implied by the model. For each fund *i* at time *t*, I generate predicted monthly alphas for the subsequent 12-month period (*t* + 1, ..., *t* + 12). Using these predictions, I sort funds into decile portfolios each month and construct equally-weighted portfolios that are rebalanced monthly. I then evaluate the performance of these portfolios by calculating their average monthly Carhart four-factor alphas over the following year.

As shown in Column 1 of Table 4, the diagnostic portfolio constructed by taking the top decile minus the bottom decile exhibits a monthly alpha spread of 23.24 basis points, equivalent to an annualized alpha spread of 2.8%. This difference is statistically significant at the 1% level, confirming the model's ability to identify meaningful performance differentials among funds. While shorting mutual funds is generally not feasible, this spread serves as a robust measure of the model's cross-sectional ranking power.

To examine whether this performance advantage persists over longer horizons, I maintain the same portfolio allocations and track their performance over the subsequent five years. Remarkably, the top-minus-bottom spread continues to reflect both persistent cross-sectional differentiation and statistically significant abnormal performance for up to four years, with a monthly alpha spread of 22.21 basis points in the fourth year. This persistent dispersion suggests that the TFT model successfully identifies funds with sustainable skill rather than merely capturing short-term performance fluctuations.

5. Model explanation

While Section 4 validates the predictive performance of the TFT model, this section uses its interpretable outputs to identify what drives that performance. A key advantage of the TFT is that it provides transparent information about how it weights variables and historical periods through its variable selection network and self-attention mechanisms.

5.1. Variable importance analysis

The TFT model allows for opening the model black box with its explainable output—variable importance measures from the variable selection network (described in Section 3.4). The variable importance $w_{i,j,t}$, from Eq. (2), represents the weight of each variable *j* of fund *i* at time *t*. This structure provides a dynamic interpretation of relative variable importance for each fund at different times.

I first average $w_{i,j,t}$ across funds and overtime to understand the overall variable importance. Fig. 2 reports the rank of overall variable importance in the model. Consistent with the time-series design of the TFT model, the historical alpha is the most prominent variable, accounting for 28.8% of the model's decision weight. This suggests that a fund's own return history and its correlation with other variables conveys the most information in predicting future performance. This finding aligns with the notion that funds' past performance patterns, when properly interpreted through a nonlinear model, contain valuable signals about their investment strategies and skill.

Following historical alpha, several macroeconomic and market variables emerge as the next most important predictors. These include equity market returns (particularly the S&P 500 index), fixed income spread metrics (default spread and term spread), and monetary policy indicators (inflation and Treasury yields). The prominence of these variables suggests that the model captures when funds are more exposed to, or benefit from, particular information environments.

Table 4
Persistence of fund performance.

	Prediction horizon				
	0–12 months	12–24 months	24–36 months	36–48 months	48–60 months
Bottom	–25.46	–19.16	–22.10	–21.44	–9.10
2	–20.90	–17.91	–18.98	–18.60	–15.20
3	–16.90	–16.35	–20.06	–20.26	–13.71
4	–18.09	–14.19	–20.30	–17.91	–13.30
5	–10.92	–12.77	–14.25	–15.47	–7.88
6	–12.86	–12.55	–11.80	–12.84	–13.83
7	–11.51	–12.66	–9.68	–11.75	–13.25
8	–7.75	–10.96	–7.59	–8.65	–8.49
9	–6.16	–6.65	–4.79	–4.83	–2.97
Top	–1.79	–2.18	2.43	–0.65	–5.14
Top–Bottom	23.67***	16.97***	24.53***	20.79***	3.96
<i>t</i> -Statistic	(9.68)	(6.14)	(8.00)	(5.81)	(0.92)

This table reports the persistence analysis results of the TFT model. It presents monthly Carhart four-factor alphas (in basis points) for decile portfolios formed based on the TFT model's predictions across different time horizons: (i) 0–12 months, (ii) 12–24 months, (iii) 24–36 months, (iv) 36–48 months, and (v) 48–60 months. The results reflect 84 individual monthly observations over the 2012m1–2019m12 out-of-sample period. The *t*-statistics, in parentheses, are based on standard errors clustered by funds. ***, **, * denote statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

5.2. Seasonal patterns in historical alpha importance

I continue to examine the variable importance of the most important variable: historical alpha (denoted as *Alpha Importance*). When analyzing *Alpha Importance* over 12-month windows, I discover a striking seasonal pattern that appears to be aligned with corporate earnings cycles. Within the sample, I aggregate *Alpha Importance* across all funds and calendar months of the year and then aggregate the number of earnings-related announcements of public companies in each calendar month of the year.

Fig. 3 plots the relationship between the one-month-ahead *Alpha Importance* and the number of earnings announcements in the month. The figure reveals that the pattern of one-month-ahead alpha importance closely tracks the number of earnings announcements throughout the year, with both series showing notable peaks in the months following major earnings seasons. This alignment suggests that the model effectively identifies periods when fund returns are most sensitive to fundamental information flows in the market.

This finding extends the work of Pinnuck (2005), who found that earnings information explains approximately 25% of a mutual fund's average monthly abnormal performance. My results suggest that skilled fund managers may be particularly adept at processing earnings-related information and that the TFT model can identify this temporal pattern in historical returns. The seasonal variation in variable importance highlights how the model adapts its focus based on the information content of different calendar periods, rather than applying a static weighting scheme.

5.3. Attention analysis across market regimes

Beyond variable importance, the TFT model's self-attention mechanism provides additional insights into which historical time periods contain the most valuable information for predicting future returns. The attention mechanism, described in Eq. (4), assigns weights to different historical observations based on their relevance for forecasting.

Fig. 5 visualizes the average attention weights assigned by the model across the entire sample period. The most striking pattern in this visualization is the model's tendency to assign significantly higher attention weights to observations during financial crisis periods (highlighted in blue) compared to normal market conditions. This suggests that fund behavior during crisis periods reveals particularly valuable information about their risk management capabilities, investment strategies, and potential future performance.

Quantitatively, the attention weights during NBER-defined recession periods are approximately 46% higher than in non-crisis periods.

This pattern remains consistent across different fund categories and persists even when controlling for other factors like fund size, style, and fund flow.

The heightened attention on crisis periods suggests that extreme market conditions may provide particularly valuable information about fund managers' true capabilities and investment approaches. These crisis periods represent environments where the differences in fund strategies become more pronounced and consequential. When market conditions shift dramatically, fund managers must adapt their strategies or reveal limitations in their approach. This finding aligns with the central theme of my subsequent analysis: fund managers possess specific skills that emerge under particular market conditions, and these skills become most visible during periods of significant market stress.

6. Model-based conditional performance persistence

The model's recognition of the conditional nature of fund skill helps explain its ability to detect persistent patterns in fund performance. This insight provides a new perspective on one of the most fundamental questions in the mutual funds literature: whether mutual fund performance is persistent. The most influential paper on this subject, Carhart (1997), uses the net alpha earned by investors to measure managerial skill and concludes that there is no evidence of skilled or informed mutual fund managers. Fama and French (2010) use alpha measures to obtain a cross-sectional distribution of managerial talent and find evidence of inferior and superior performance only in the extreme tails of the cross-section of mutual fund performance. More recently, Berk and van Binsbergen (2015) propose a new value-generated measure for mutual fund skill and find that performance persistence exists over longer time horizons. The interpretable features of my model allow me to examine performance persistence through a more nuanced lens—specifically when predictive variables are most informative.

6.1. Variable importance conditional alpha

The model's predictive power derives from its ability to capture fund skills embedded in dynamic strategies. To better understand how mutual fund managers adopt dynamic investment strategies, I construct model-based conditional skill measures that depend on the inferred informativeness of macroeconomic and fundamental variables. Specifically, I measure the abnormal performance during periods when the model identifies particular variables as highly informative.

For each fund *i*, I calculate the variable importance conditional alpha (V.I. conditional alpha) as the average Carhart four-factor alpha during variable-specific high-informativeness periods over the past five

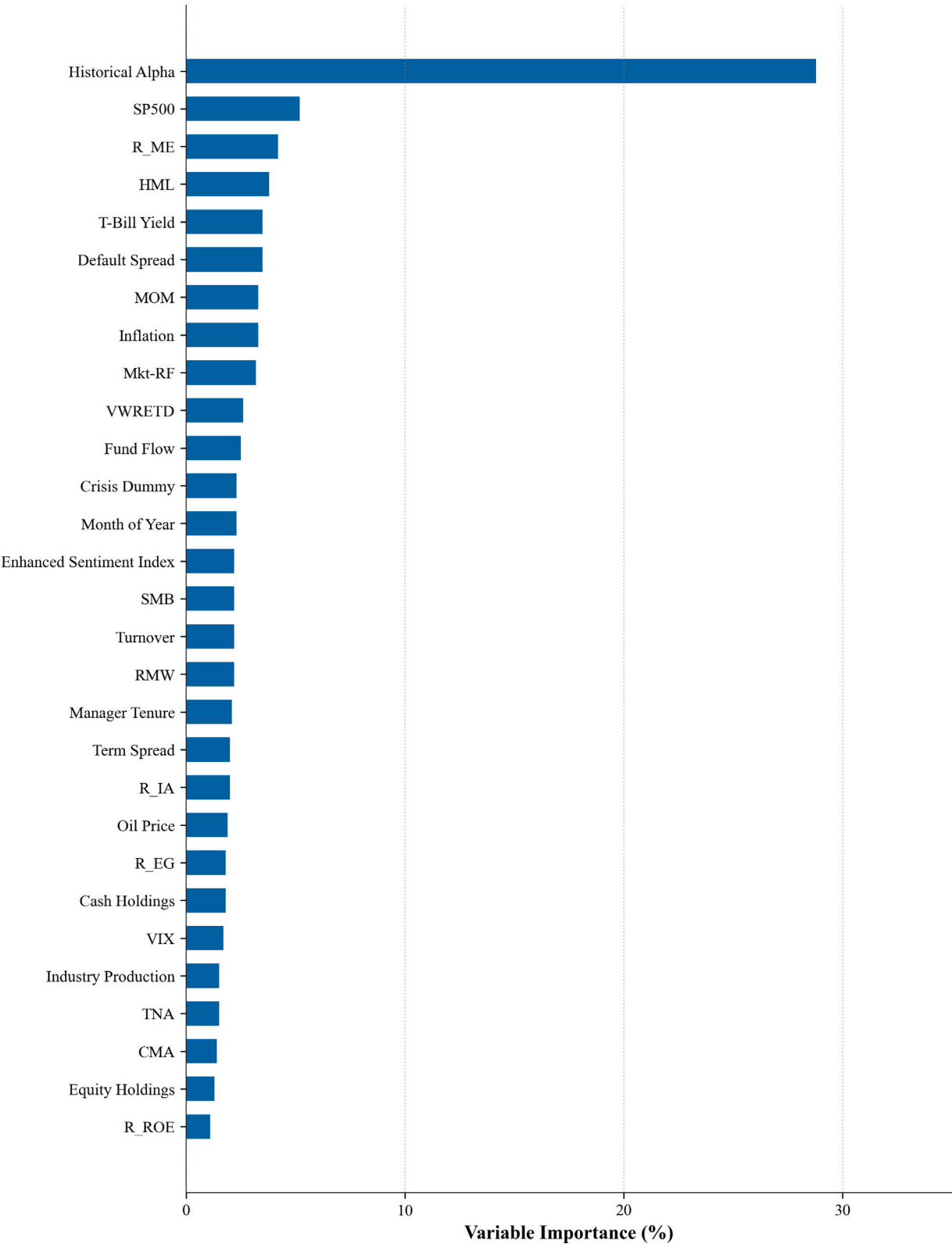


Fig. 2. Variable importance in the temporal fusion transformer model. This figure displays the relative importance of all variables in the TFT model for the period from January 1990 to December 2009. For each variable, the importance weights are first averaged across all funds and then across all months, with results expressed in percentage points. All variables are defined in [Appendix A](#).

years. A high-informativeness period (low-informativeness period) is defined as months when the variable importance assigned by the model to a particular variable for a fund exceeds (falls below) that

of 80 percent of other funds in the same month. Importantly, these variable importance weights are obtained strictly from the in-sample model estimation (1990–2009), ensuring no information leakage or

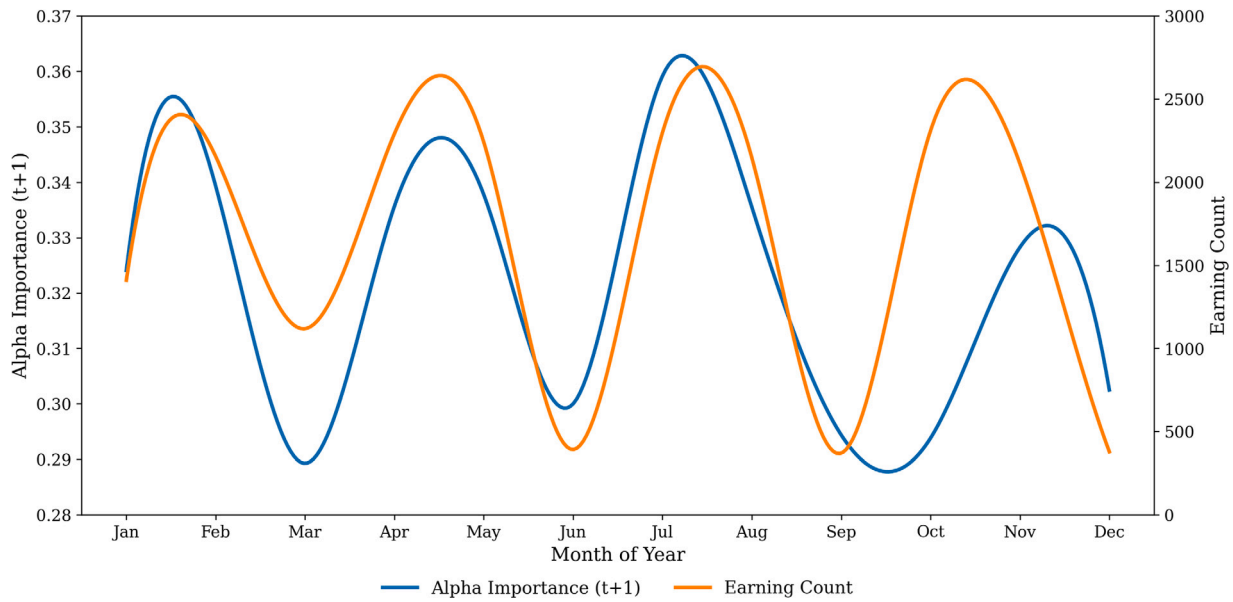


Fig. 3. Variable importance of returns and frequency of earnings announcements.

This figure plots the relationship between the one-month ahead variable importance of mutual fund risk-adjusted returns (alphas) and the number of earnings announcements in the corresponding month. The variable importance of mutual fund alphas is first averaged across all funds and then averaged across all years in the sample for each calendar month. Blue-shaded regions indicate NBER-defined economic recession periods. The higher attention weights during crisis periods suggest these timeframes contain particularly valuable information for predicting future fund performance.

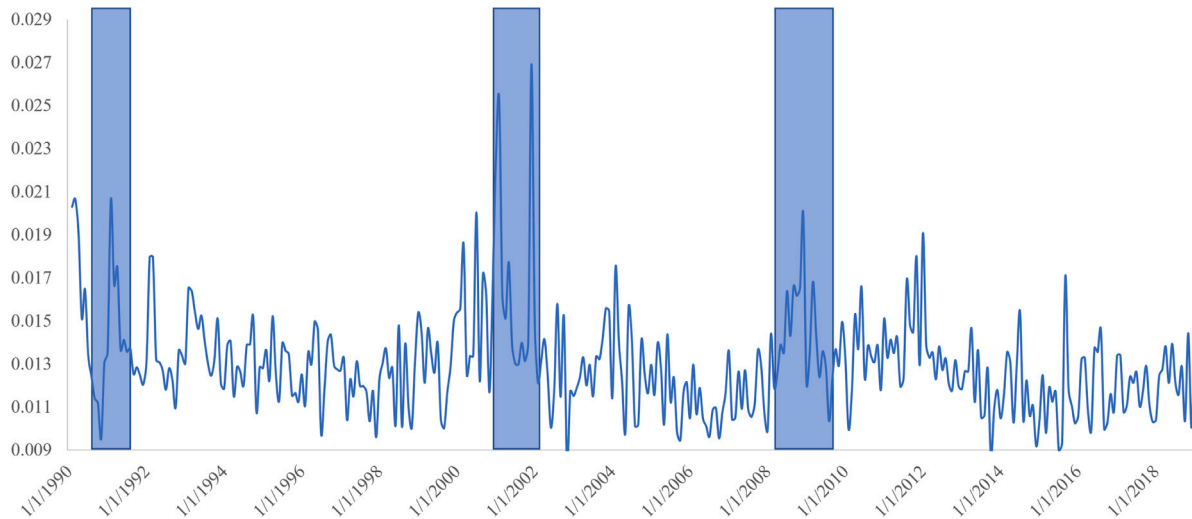


Fig. 4. Conditional performance persistence.

This figure plots confidence intervals of subsequent one-year performance based on V.I. conditional alpha rankings during high and low informativeness periods for both macroeconomic and fundamental variables. V.I. conditional alpha is calculated as the average Carhart's four-factor alpha during periods when a variable is highly informative over the past five years. High (low) informativeness is defined as months when the variable importance for a specific fund exceeds (falls below) that of 80 percent of other funds in the same month. For each calendar year from 2010 to 2019, funds are sorted into quintile portfolios based on their V.I. conditional alpha. Future performance is then separated into high and low informativeness periods. The x-axis represents the V.I. conditional alpha quintile rank, while the y-axis shows the Carhart's four-factor alpha.

look-ahead bias in the construction of the conditional alphas for the out-of-sample evaluation period. The conditional variables fall into two distinct categories: (1) macroeconomic variables, including market return, inflation, term spread, and default spread, and (2) fundamental variables, including fund past returns and month-of-year indicators.

Intuitively, this construction defines a fund-specific *informativeness state*: a month in which the model places unusually high weight on a particular variable for that fund. The central question is whether funds that historically earn high alpha in a given informativeness state

continue to earn high alpha when the same type of informativeness state reappears in the future.

Table 5 reports the correlation matrix of V.I. conditional alphas across different macroeconomic and fundamental variables. The generally low correlations among these measures suggest that they capture distinct dimensions of fund skill. For example, the measure derived from month-of-year, which captures the seasonality of fundamental information, exhibits near-zero correlation with other measures. Some measures show moderate correlation where their underlying

Table 5
Correlation of V.I. Conditional Alpha.

	Market return	Inflation	Term spread	Default spread	Alpha importance	Month of year
Market return	1.00					
Inflation	0.22	1.00				
Term spread	−0.43	−0.13	1.00			
Default spread	0.14	0.23	−0.05	1.00		
Alpha importance	−0.15	−0.09	0.25	0.00	1.00	
Month of year	0.06	0.01	0.04	−0.05	−0.04	1.00

This table reports the correlations among V.I. conditional alphas based on different macroeconomic and fundamental variables. For each variable, the V.I. conditional alpha is calculated as the average Carhart's four-factor alpha during periods when that variable is highly informative over the past five years. A period is classified as highly informative (low-informativeness) when the variable importance assigned by the model to a specific variable for a fund exceeds (falls below) that of 80 percent of other funds in the same month.

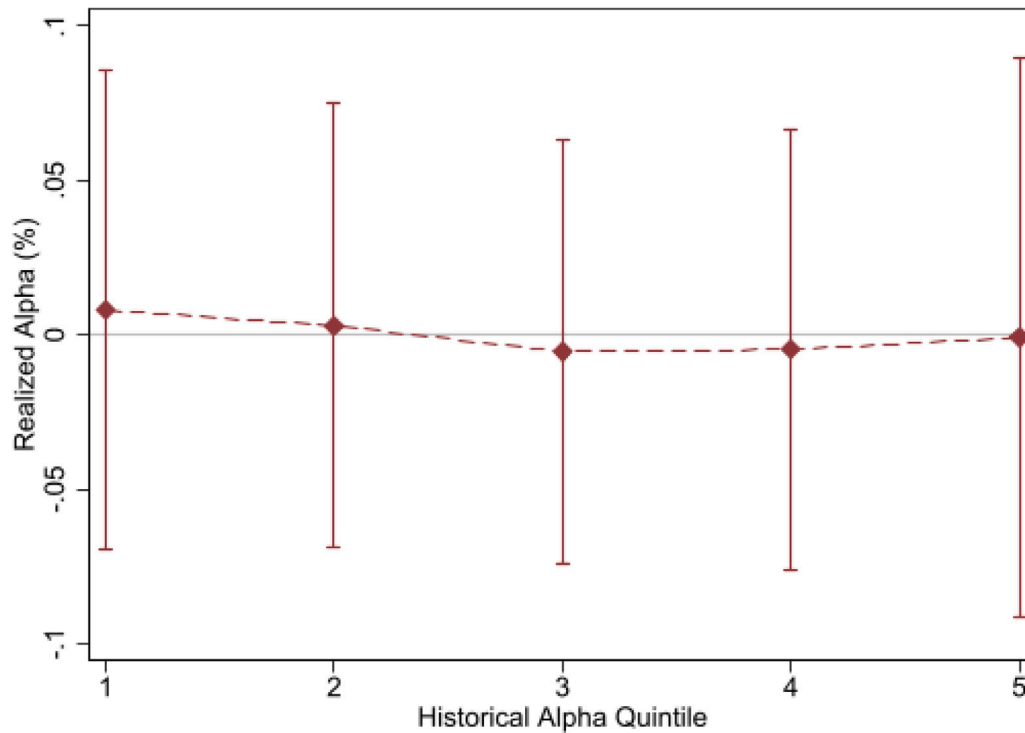


Fig. 5. Time series of model attention weights across sample period.

This figure displays the average attention weights assigned by the Temporal Fusion Transformer model across the sample period from 1990 to 2019. For each calendar month, the attention value represents the average weight the model assigned to that specific month when making predictions, aggregated across all funds and forecast horizons. Blue-shaded regions indicate NBER-defined economic recession periods. The higher attention weights during crisis periods suggest these timeframes contain particularly valuable information for predicting future fund performance.

macroeconomic conditions overlap—for instance, the correlation between inflation-based and market return-based measures is 0.27.

6.2. Empirical evidence of conditional performance persistence

I first examine performance persistence without conditioning on model-identified informativeness periods. Following Carhart (1997), I form portfolios of mutual funds based on lagged 12-month Carhart's four-factor alphas and estimate future 12-month alphas for these portfolios. On January 1 of each year, I form five equal-weighted portfolios of mutual funds using historical alphas. As shown in Fig. 6, these portfolios demonstrate minimal variation in mean alphas, consistent with Carhart (1997)'s finding that performance persistence largely disappears after controlling for momentum effects.

I then reform five equal-weighted portfolios using the V.I. conditional alphas based on different macroeconomic and fundamental variables over the past five years. The results, presented in Fig. 4,

reveal that conditional skill measures predict fund performance in future periods when the conditioning variables are highly informative. Mean alphas increase monotonically and significantly during high-informativeness periods (when the model recognizes similar informative conditions in the future). This pattern appears consistently across both macroeconomic and fundamental information variables. However, no similar pattern emerges during low-informativeness periods. These findings suggest that fund managers' skills may only manifest under specific conditions—when macroeconomic variables align with their investment strategies or when fundamental information is more readily available in the market.

To formally test this hypothesis, I conduct panel regression analysis by estimating the following specification, indexed by fund(*i*)-month(*t*), with both year and style fixed effects, and controlling for *Historical Alpha*, *log(TNA)*, and *Fund Flow*:

$$Alpha_{i,t+1} = \beta V.I. Conditional Alpha_{i,t} + \gamma Control_{i,year} + \alpha_i + \alpha_t + \epsilon_{i,t} \quad (7)$$

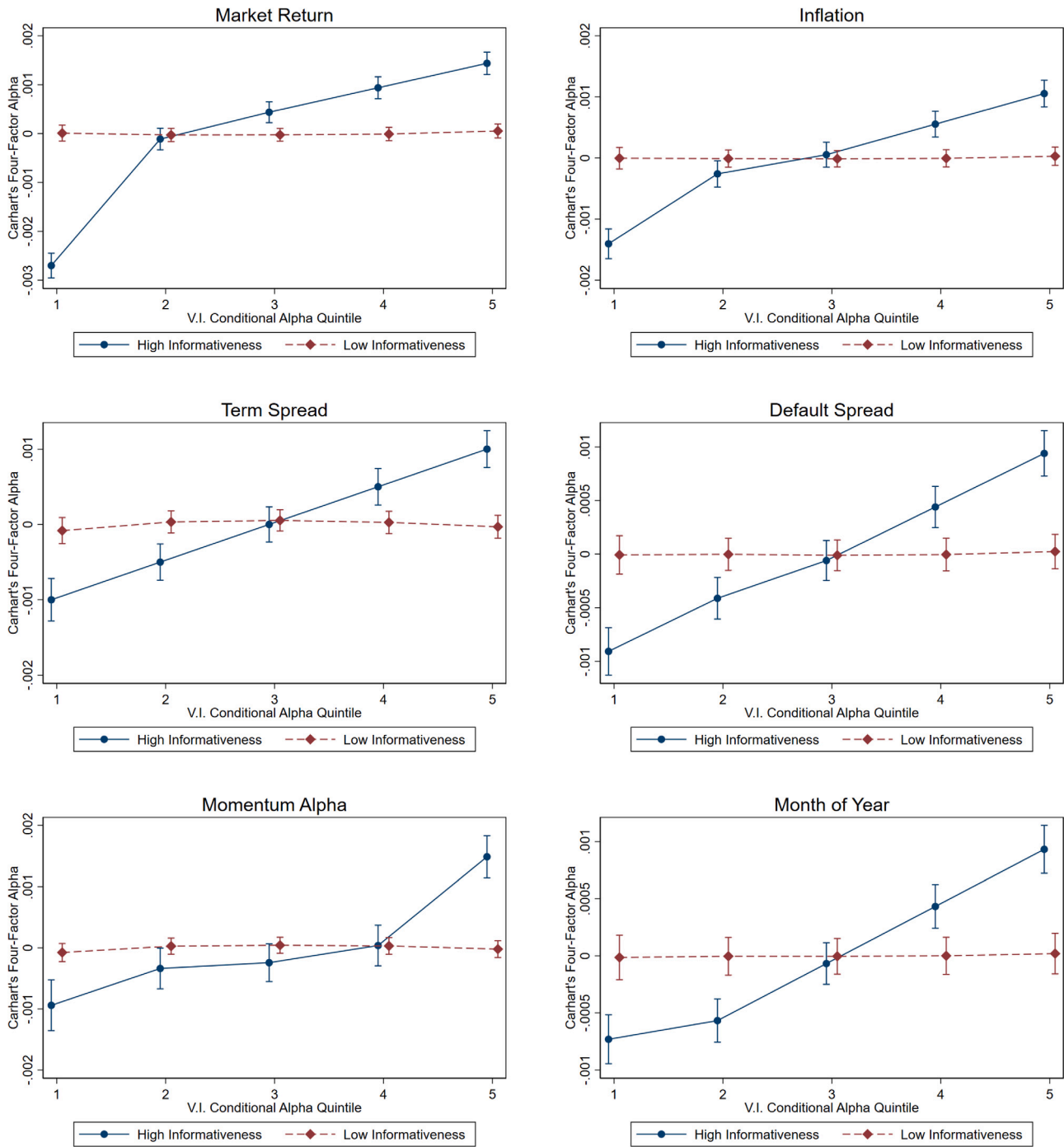


Fig. 6. Unconditional performance persistence.

This figure plots the confidence intervals of subsequent one-year performance by quintile portfolios ranked on historical alpha. For each calendar year from 2010 to 2019, funds are sorted into quintile portfolios based on their average Carhart's four-factor alpha over the past 12 months.

Table 6 presents the regression results. The coefficient of *Historical Alpha* in column one from Panel A is statistically significant but economically modest. In contrast, the coefficients of *V.I. Conditional Alpha* are both economically and statistically significant during high-informativeness periods for both fundamental and macroeconomic variables. Notably, *Historical Alpha* becomes insignificant in these specifications. In low-informativeness periods, the predictive relationship attenuates: when the relevant information environment is absent, strategies that add value in high-informativeness states do not translate into positive future alpha. This asymmetry is consistent with state-dependent skill rather than unconditional persistence.

6.3. Conditional performance persistence in long term

To determine whether conditional performance persistence extends beyond the short term, I examine the predictive power of V.I. conditional alphas over multiple horizons: (i) 12–24 months, (ii) 24–36 months, and (iii) 36–48 months. Table 7 presents the coefficients and *t*-statistics from regressions of multi-horizon future fund alphas on V.I. conditional alphas during high-informativeness periods, following Eq. (7). The results demonstrate that conditional performance persistence remains significant for up to 48 months. This suggests that fund managers apply similar investment strategies when specific information

Table 6
Conditional performance persistence.

Panel A: Fundamental conditions								
Dependent variable	(1)	(2)		(3)		(4)		
	Alpha importance		Future alpha		Month of the year			
	High	Low	High	Low	High	Low	High	Low
<i>V.I. Conditional Alpha</i>	0.242*** (10.23)	−0.023*** (−2.94)	0.105*** (5.02)	0.001 (0.17)				
<i>Historical Alpha</i>	−0.027 (−0.66)	0.116*** (6.42)	−0.004 (−0.13)	0.162*** (8.52)				
<i>Log (TNA)</i>	0.003*** (12.50)	0.000*** (12.76)	−0.000 (−1.18)	−0.000* (−1.72)				
<i>Flow</i>	−0.031*** (−9.90)	−0.003*** (−2.82)	0.004* (1.72)	−0.000 (−0.20)				
Time fixed effect	Yes	Yes	Yes	Yes				
Style fixed effect	Yes	Yes	Yes	Yes				
Observations	18,348	62,847	18,462	64,184				
adjust R-squared	0.204	0.0808	0.170	0.0504				

Panel B: Macroeconomics conditions								
Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Market return		Term spread		Inflation		Default spread	
	High	Low	High	Low	High	Low	High	Low
<i>V.I. Conditional Alpha</i>	0.530*** (24.66)	−0.082*** (−7.57)	0.439*** (14.36)	−0.065*** (−6.67)	0.273*** (12.46)	−0.001 (−0.10)	0.118*** (3.67)	−0.005 (−0.62)
<i>Historical Alpha</i>	0.057 (1.13)	0.129*** (6.13)	0.058 (1.00)	0.158*** (7.01)	−0.000 (−0.00)	0.167*** (8.60)	0.063** (2.35)	0.166*** (8.96)
<i>Log (TNA)</i>	−0.006*** (−20.34)	0.000*** (11.94)	0.004*** (16.21)	0.000*** (5.63)	−0.006*** (−34.54)	0.000*** (2.72)	−0.002*** (−18.24)	0.000 (0.37)
<i>Flow</i>	−0.015*** (−4.77)	−0.010*** (−9.86)	−0.003 (−1.63)	0.009*** (7.41)	−0.012*** (−5.01)	−0.005*** (−6.00)	−0.005 (−1.35)	0.001 (0.88)
Time fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Style fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	18,569	70,509	18,509	64,584	18,375	67,022	18,543	67,227
adjust R-squared	0.231	0.0641	0.204	0.0942	0.239	0.0563	0.175	0.0533

This table reports the regression of future fund alpha on V.I. conditional alpha during high and low informativeness periods. V.I. conditional alpha is calculated as the average Carhart's four-factor alpha during periods when a specific variable is highly informative over the past five years. A high-informativeness period (low-informativeness period) is defined as months when the variable importance assigned by the model to a particular variable for a fund exceeds (falls below) that of 80 percent of other funds in the same month. Control variables include historical alpha, fund size, and fund flow. The *t*-statistics, in parentheses, are based on standard errors clustered by funds. ***, **, * denote statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

types become informative again in the future, consistent with theories of limited attention and specialization in investment management.

6.4. Distinguishing model-based from direct macroeconomic conditioning

An alternative explanation for these findings could be that fund performance simply persists during periods when actual macroeconomic variables reach certain thresholds—for instance, during high inflation periods. To differentiate between model-identified informativeness and raw macroeconomic conditions, I augment Eq. (7) by adding *Macro Conditional Alpha* as a control variable. This measure is calculated as the average Carhart four-factor alpha during periods when a macroeconomic variable exceeds its 80th percentile threshold in the sample from 1990m1 to 2019m12.

Table 8 presents these results. While *Macro Conditional Alpha* demonstrates some statistically significant predictive power, its effects are inconsistent in direction. For example, funds that perform well when market returns are high historically tend to perform worse when market returns are high in the future. Overall, these findings suggest that fund performance persists specifically when certain information types are particularly informative to funds' investment processes, rather than

simply during periods when macroeconomic variables reach certain thresholds.

Table 8 helps disentangle two channels: (i) coarse macro regimes that mechanically affect fund opportunities (captured by Macro Conditional Alpha), and (ii) fund-specific, model-identified informativeness states that govern when a manager's strategy is most effective (captured by V.I. Conditional Alpha). Once macro tailwinds/headwinds are controlled for, Historical Alpha can regain explanatory power because it contains residual fund-specific persistence not fully summarized by macro-threshold regimes. Importantly, V.I. Conditional Alpha remains strongly predictive in high-informativeness states, reinforcing the interpretation that persistence is primarily conditional and state-dependent.

6.5. Conditional performance persistence and fund size

Berk and Green (2004) hypothesize that fund performance persistence is difficult to maintain, especially as funds grow larger, due to diminishing returns to scale in the mutual fund industry. To test whether similar scale effects apply to V.I. conditional performance, I divide the sample into quintiles based on fund size and examine the

Table 7
Conditional performance persistence in long term.

Market return	Term spread	Inflation	Default spread	Alpha importance	Month of the year
High	High	High	High	High	High
12–24 Months					
0.430*** (20.29)	0.337*** (12.31)	0.196*** (10.26)	0.097*** (3.33)	0.242*** (10.23)	0.105*** (5.02)
24–36 Months					
0.339*** (13.50)	0.293*** (10.54)	0.169*** (8.96)	0.101*** (3.81)	0.201*** (8.36)	0.098*** (5.21)
36–48 Months					
0.300*** (15.04)	0.261*** (10.19)	0.127*** (8.23)	0.091*** (4.59)	0.152*** (6.71)	0.073*** (4.01)

This table reports the coefficients and *t*-statistics from regressions of future fund alphas on V.I. conditional alphas during high-informativeness periods across multiple prediction horizons: (i) 12–24 months, (ii) 24–36 months, and (iii) 36–48 months. V.I. conditional alpha is calculated as the average Carhart's four-factor alpha during variable-specific high-informativeness periods over the past five years. A high-informativeness (low-informativeness) period is defined as months when the variable importance assigned by the model to a particular variable for a fund exceeds (falls below) that of 80 percent of other funds in the same month. Control variables include historical alpha, fund size, and fund flow. The *t*-statistics, in parentheses, are based on standard errors clustered by funds. ***, **, * denote statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

Table 8
Conditional performance persistence on macro conditional alpha.

Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Future alpha							
Conditions	Market return		Term spread		Inflation		Default spread	
	High	Low	High	Low	High	Low	High	Low
V.I. Conditional Alpha	0.518*** (23.76)	−0.077*** (−7.24)	0.422*** (13.65)	−0.063*** (−6.45)	0.274*** (12.49)	−0.006 (−0.75)	0.110*** (3.40)	−0.006 (−0.83)
Macro Conditional Alpha	−0.196* (−1.84)	−0.052 (−1.00)	−0.253** (−2.48)	−0.083 (−1.46)	0.037 (0.35)	0.420*** (5.42)	0.132** (2.13)	0.149*** (2.94)
Historical Alpha	0.060** (2.00)	0.109*** (4.92)	0.062* (1.74)	0.143*** (6.19)	−0.000 (−0.00)	0.128*** (6.65)	0.051* (1.88)	0.119*** (6.16)
Log (TNA)	−0.005*** (−18.51)	−0.000** (−2.22)	0.004*** (14.55)	−0.003*** (−28.87)	−0.005*** (−31.01)	0.000 (1.65)	−0.002*** (−15.76)	−0.001*** (−13.64)
Flow	−0.015*** (−4.79)	−0.009*** (−9.84)	−0.004** (−2.50)	0.009*** (7.95)	−0.012*** (−4.38)	−0.005*** (−6.03)	−0.006* (−1.67)	0.001 (1.16)
Time fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Style fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	18,890	73,180	18,787	67,164	18,688	69,422	18,829	69,665
adjust R-squared	0.241	0.0632	0.212	0.0935	0.242	0.0564	0.172	0.0523

This table reports the regression of future fund alpha on V.I. conditional alpha while controlling for macro conditional alpha. V.I. conditional alpha is calculated as the average Carhart's four-factor alpha during variable-specific high-informativeness periods over the past five years. A high-informativeness period (low-informativeness period) is defined as months when the variable importance assigned by the model to a particular variable for a fund exceeds (falls below) that of 80 percent of other funds in the same month. The macro conditional alpha is calculated as the average Carhart's four-factor alpha during periods when a macroeconomic variable exceeds its 80th percentile threshold in the sample from 1990m1 to 2019m12. Regressions are presented separately for high-informativeness and low-informativeness periods. Control variables include historical alpha, fund size, and fund flow. The *t*-statistics, in parentheses, are based on standard errors clustered by funds. ***, **, * denote statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

predictive power of V.I. conditional alphas within each quintile using Eq. (7).

Table 9 presents these results. The evidence regarding scale effects is mixed. For measures such as Term Spread and Alpha Importance, the predictive power of V.I. conditional alpha diminishes for the largest mutual funds (Group 5), consistent with the diminishing returns-to-scale hypothesis. However, for other signals like Inflation and Month of the Year, the predictive relationship remains robust or even strengthens among the largest funds. To formally test whether these differences are statistically significant, I report the coefficient difference between Group 5 and Group 1 using an interaction regression framework. The results (bottom row of Table 9) show that most differences are not statistically significant, with the exception of Month of Year (*t* = 2.02). This heterogeneity suggests that while scale may hinder the exploitation of general inefficiencies, large funds may possess superior resources to capitalize on specific complex signals, offsetting the disadvantages of size.

6.6. Conditional performance persistence and manager turnover

To further validate that the model captures fund-level skill rather than just manager-specific traits, I examine the impact of manager turnover on conditional performance persistence. Following Khorana (2001) and Chevalier and Ellison (1999), manager turnover is measured as either a dummy variable indicating any manager change or a continuous turnover rate over a rolling six-year window. As shown in the Online Appendix (Table OA.2), the interaction between V.I. Conditional Alpha and manager turnover is statistically insignificant across most specifications. This indicates that the conditional skill identified by the model is robust to changes in the management team. It suggests that the “skill” is likely capturing a fund-level strategy or institutional capability that persists even when individual managers change.

7. Conclusion remarks

This paper proposes a new framework for understanding mutual fund performance through the lens of temporal and state-dependent

Table 9
Conditional performance persistence by fund size.

Group	Market return	Inflation	Term spread	Default spread	Alpha importance	Month of year
	High	High	High	High	High	High
5	0.486*** (10.08)	0.319*** (6.43)	0.311*** (5.24)	0.109** (2.40)	0.238*** (5.28)	0.237*** (4.62)
4	0.512*** (10.71)	0.244*** (5.30)	0.476*** (9.73)	0.156 (1.51)	0.290*** (5.10)	0.119*** (2.90)
3	0.513*** (14.45)	0.334*** (6.66)	0.453*** (10.66)	0.085* (1.94)	0.349*** (6.94)	0.134*** (3.19)
2	0.501*** (9.65)	0.229*** (5.01)	0.307*** (4.36)	0.094* (1.88)	0.331*** (7.46)	0.072 (1.62)
1	0.523*** (12.44)	0.230*** (5.58)	0.410*** (6.85)	0.067 (1.46)	0.309*** (7.28)	−0.011 (−0.19)
Diff (5–1)	−0.037 (−0.63)	0.089 (0.96)	−0.099 (−0.23)	0.042 (0.39)	−0.071 (−0.57)	0.248** (2.02)

This table reports the regression of future fund alpha during high-informativeness periods on V.I. conditional alpha, grouped by fund size. V.I. conditional alpha is calculated as the average Carhart's four-factor alpha during variable-specific high-informativeness periods over the past five years. A high-informativeness (low-informativeness) period is defined as months when the variable importance assigned by the model to a particular variable for a fund exceeds (falls below) that of 80 percent of other funds in the same month. Control variables include historical alpha, fund size, and fund flow. The bottom row reports the difference between Group 5 and Group 1 coefficients, tested using a pooled regression with an interaction term between V.I. Conditional Alpha and a Large fund dummy (Group 5 = 1, Group 1 = 0). The *t*-statistics, in parentheses, are based on standard errors clustered by funds. ***, **, * denote statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

dynamics. Rather than focusing on tradable portfolio strategies, the analysis uses machine learning to identify when managerial skill becomes informative and when it persists. The findings offer several implications for mutual fund performance evaluation.

First, the Temporal Fusion Transformer model demonstrates superior predictive power compared to traditional econometric and machine learning approaches. A diagnostic long-short portfolio based on the model's predictions exhibits a statistically significant annualized alpha spread of 2.8%, with dispersion persisting for up to four years.

Second, the model's interpretable features reveal that predictive signals are state-dependent. Historical fund returns, macroeconomic variables, and market states contribute to forecast accuracy, and the model assigns significantly higher attention weights to observations during financial crises, suggesting that stress periods are particularly informative about future performance.

Third, conditional skill measures constructed from the model's variable-importance weights indicate that some funds earn alpha primarily when particular information types are informative. For example, funds with high inflation-state conditional alpha outperform specifically when inflation is highly informative. These conditional effects persist for up to four years.

These findings imply that conventional persistence tests can understate skill by averaging across states in which a fund's strategy is not informative. Conditioning on model-identified informativeness states helps reconcile the difficulty of detecting unconditional persistence with evidence of persistent, state-dependent skill.

This research contributes to the growing literature applying machine learning to mutual fund analysis by emphasizing temporal dynamics and fund-specific informativeness states, complementing work that focuses primarily on cross-sectional selection using characteristics.

Future research could extend this framework to other information environments and to alternative definitions of informativeness states, and could use interpretability outputs to construct benchmarks that explicitly account for state-dependent skill.

CRediT authorship contribution statement

Norman (Xuxi) Guo: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Appendix A. Definitions of variables

Variable	Definition
<i>Alpha</i>	The Fama–French–Carhart four-factor alpha is the intercept of the rolling window regression of the monthly net return during 24 months on Mktrf, SMB, HML, and UMD factors, expressed in percentage points.
<i>TNA, Log(TNA)</i>	TNA is a fund's TNA (\$mm) prior to month <i>t</i> . Log(TNA) is natural logarithm of a fund's TNA.
<i>Flow</i>	The monthly flow for a fund in month <i>t</i> -1, calculated as $Flow = TNA_{i,t} / TNA_{i,t-1} - 1 - r_{i,t}$, where $r_{i,t}$ is the net return in the prior month, expressed in percentage points.
<i>Expense</i>	The most recent expense ratio prior to month <i>t</i> .
<i>Turnover</i>	The most recent turnover ratio prior to month <i>t</i> .
<i>Load</i>	Dummy variable if funds have load.
<i>Return Gap</i>	The Return Gap measure from Kacperczyk, Sialm, and Zheng (2008).
<i>Cash Holdings</i>	The most recent amount of fund invested in cash prior to month <i>t</i> , expressed in percentage.
<i>Management Fee</i>	The most recent management fees prior to month <i>t</i> .
<i>Manager Tenure</i>	The number of months since a portfolio manager is hired. If there are multiple managers for a fund, the longest tenure is used.
<i>Manager Turnover</i>	The rate of portfolio manager turnover for a fund in a given year.
<i>Alpha Mean</i>	Mean of the alpha in a fund-year.
<i>Alpha Std</i>	Standard deviation of the alpha in a fund-year.
<i>Industry production</i>	Percentage change of industry production index (INDPRO) from year ago.

<i>Inflation</i>	Percentage change of consumer price index for all urban consumers (CPIAUCSL) from year ago.
<i>Oil price</i>	Percentage change of crude oil prices:West Texas Intermediate (WTI) from year ago.
<i>T-Bill yield</i>	Percentage change of 3-month treasury bill (TB3MS).
<i>Term spread</i>	Percentage change of the difference between 10-year treasury (GS10) and 3-month treasury bill (TB3MS).
<i>Default spread</i>	Percentage change of the difference between Baa corporate bond yield (BAA) and Aaa corporate bond yield (AAA).
<i>Crisis Dummy</i>	Crisis dummy defined by NBER.
<i>VIX</i>	Percentage change of The CBOE volatility index.
<i>VWRETD</i>	Return of total return value-weighted index from CRSP.
<i>SP500</i>	S&P 500 index return.
<i>Mkt-RF</i>	Market excess return.
<i>SMB</i>	Size factor return in Fama–French five-factor (FF5) Model.
<i>HML</i>	Value Factor return in FF5 Model.
<i>RMW</i>	Profitability Factor return in FF5 Model.
<i>CMA</i>	Investment Factor return in FF5 Model.
<i>R_ME</i>	Value Factor return in Hou, Xue, and Zhang (2015) q-factor (q5) Model.
<i>R_IA</i>	Investment factor return in q5 Model.
<i>R_ROE</i>	Equity factor return in q5 Model.
<i>R_EG</i>	Expected growth factor return in q5 Model.
<i>Enhanced Sentiment</i>	The enhanced investor sentiment index from Baker and Wurgler (2006) and Ung, Gebka, and Anderson (2024) .
<i>Announcement Count %</i>	Percentage of the number of the earning announcement of a fund's holding in a month as that number of a year.
<i>Alpha Importance</i>	Variable importance of the historical alpha in the model.
<i>Macro Importance</i>	Sum of the variable importance of macroeconomics in the model. The variables include <i>Industry Production</i> , <i>Inflation</i> , <i>Oil Price</i> , <i>T-Bill</i> , <i>Term Spread</i> , <i>Default Spread</i> , <i>Crisis Dummy</i> , <i>VIX</i> , <i>VWRETD</i> , <i>SP500</i> , <i>Mkt-RF</i> , <i>SMB</i> , <i>HML</i> , <i>RMW</i> , <i>CMA</i> , <i>R_ME</i> , <i>R_IA</i> , <i>R_ROE</i> , <i>R_EG</i>

Appendix B. Detailed explanation of temporal fusion transformer (TFT)

This appendix provides a comprehensive mathematical formulation and detailed architectural description of the Temporal Fusion Transformer (TFT) model as proposed by [Lim et al. \(2021\)](#). The TFT architecture is designed for multi-horizon forecasting tasks where diverse input types and long-term dependencies complicate prediction. The model introduces five integrated components that together deliver both high accuracy and interpretability.

B.1. Variable selection module

At each time step t , the model receives a set of observed and known input variables $\mathbf{x}_t = [x_t^{(1)}, x_t^{(2)}, \dots, x_t^{(m)}]$. Each variable $x_t^{(j)}$ is first transformed into a d -dimensional embedding vector $\xi_t^{(j)}$. The variable selection mechanism then determines relevance scores for each variable using a Gated Residual Network (GRN) and softmax normalization:

$$\mathbf{v}_t = \text{Softmax}(\text{GRN}_v([\xi_t^{(1)}, \dots, \xi_t^{(m)}], \mathbf{c}_s)), \quad (\text{B.1})$$

where $\mathbf{c}_s$ is a static context vector derived from static metadata S via a static covariate encoder. The input embeddings are further processed through individual GRNs and combined using the variable importance weights:

$$\tilde{\xi}_t = \sum_{j=1}^m v_t^{(j)} \cdot \text{GRN}_j(\xi_t^{(j)}). \quad (\text{B.2})$$

This mechanism is separately applied to static variables, past inputs, and known future inputs.

B.2. LSTM encoder and decoder for local temporal modeling

The temporal input embeddings $\{\tilde{\xi}_t\}$ are passed into a sequence-to-sequence architecture composed of LSTM layers:

- The encoder consumes historical data $\tilde{\xi}_{t-k:t}$, where k is the look-back window.
- The decoder consumes known future inputs $\tilde{\xi}_{t+1:t+\tau_{\max}}$, where $\tau_{\max}$ is the forecast horizon.

To incorporate static metadata, initial hidden and cell states of both encoder and decoder LSTMs are initialized using learned projections of static context vectors $\mathbf{c}_h$ and $\mathbf{c}_c$. Each temporal step yields an encoded context vector $\phi(t, n)$ which serves as the basis for further decoding and attention processing.

B.3. Interpretable multi-head attention for long-term dependencies

To model long-term temporal relationships, TFT uses a modified self-attention mechanism. For a sequence of enriched representations $\Theta(t) = [\theta(t, -k), \dots, \theta(t, \tau_{\max})]$, attention weights are computed as:

$$A(Q, K) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right), \quad (\text{B.3})$$

$$\text{Attention}(Q, K, V) = A(Q, K)V. \quad (\text{B.4})$$

To improve interpretability, TFT uses shared value projections V across all heads and aggregates attention contributions additively:

$$\tilde{H} = \frac{1}{H} \sum_{h=1}^H \text{Attention}(QW_Q^{(h)}, KW_K^{(h)}, VW_V). \quad (\text{B.5})$$

The result $\tilde{H}$ is passed through residual gating and layer normalization to produce attention-enhanced outputs.

B.4. Static enrichment and position-wise feed-forward layers

TFT enhances temporal encodings with static metadata using a static enrichment layer:

$$\theta(t, n) = \text{GRN}(\phi(t, n), \mathbf{c}_e), \quad (\text{B.6})$$

where $\mathbf{c}_e$ is a context vector from static encoders. These enriched outputs are then passed through interpretable attention, residual connections, and a position-wise feed-forward network with gated residual connections:

$$\delta(t, n) = \text{LayerNorm}(\theta(t, n) + \text{GLU}(\beta(t, n))), \quad (\text{B.7})$$

$$\psi(t, n) = \text{GRN}(\delta(t, n)), \quad (\text{B.8})$$

$$\tilde{\psi}(t, n) = \text{LayerNorm}(\psi(t, n) + \text{GLU}(\psi(t, n))). \quad (\text{B.9})$$

B.5. Quantile loss optimization for forecast intervals

TFT directly predicts multiple quantiles $\hat{y}_{t+\tau}^{(q)}$ for each forecast horizon τ and quantile level $q \in \mathcal{Q}$. The quantile loss penalizes over-

under-predictions asymmetrically:

$$QL(y, \hat{y}, q) = q \cdot (y - \hat{y})_+ + (1 - q) \cdot (\hat{y} - y)_+, \quad (B.10)$$

where $(x)_+ = \max(0, x)$. The overall loss function averaged across all quantiles and horizons is:

$$\mathcal{L}(\Omega, W) = \frac{1}{M\tau_{\max}} \sum_{i=1}^M \sum_{q \in Q} \sum_{\tau=1}^{\tau_{\max}} QL(y_{i,t+\tau}, \hat{y}_{i,t+\tau}^{(q)}, q), \quad (B.11)$$

where M is the number of training samples, and W are the trainable parameters of the model.

These components collectively enable TFT to generate multi-horizon forecasts with high accuracy and interpretability, effectively balancing model complexity with explainability—critical for financial time series modeling.

Appendix C. Model implementation and cross-validation details

This appendix provides comprehensive implementation details for the Temporal Fusion Transformer (TFT) model training, including the rolling-window cross-validation framework, hyperparameter optimization, and ensemble prediction methodology.

C.1. Rolling-window cross-validation framework

The k-fold rolling-window cross-validation scheme implements 10 distinct folds spanning the sample period. Each fold is structured with a training window of 240 consecutive months (20 years), with consecutive folds shifting the training window forward by 12 months, generating out-of-sample predictions for periods strictly following each training window. The specific configuration for each fold is:

Fold	Training period	Training QID	Test period
1	Jan 1990–Dec 2009	120–360	Jan 2010–Dec 2019
2	Jan 1991–Dec 2010	132–372	Jan 2011–Dec 2019
3	Jan 1992–Dec 2011	144–384	Jan 2012–Dec 2019
4	Jan 1993–Dec 2012	156–396	Jan 2013–Dec 2019
5	Jan 1994–Dec 2013	168–408	Jan 2014–Dec 2019
6	Jan 1995–Dec 2014	180–420	Jan 2015–Dec 2019
7	Jan 1996–Dec 2015	192–432	Jan 2016–Dec 2019
8	Jan 1997–Dec 2016	204–444	Jan 2017–Dec 2019
9	Jan 1998–Dec 2017	216–456	Jan 2018–Dec 2019
10	Jan 1999–Dec 2018	228–468	Jan 2019–Dec 2019

Note: QID represents the quarter ID encoding where $QID = (\text{Year} - 1980) \times 4 + \text{Quarter}$. Test periods begin immediately after the training window to prevent data leakage.

C.2. Cross-sectional fund split within each fold

Within each fold's 240-month training window, funds are randomly partitioned into training (80%) and validation (20%) subsets based on fund identifiers (wfcid) using a fixed random seed (seed=10) for reproducibility. This ensures: (1) training subset uses 80% of funds with all their observations over the fold's training window; (2) validation subset uses the remaining 20% of funds over the same training window; (3) no fund appears in both subsets within any fold; and (4) both subsets span the fold's entire training period, exposing the model to all market regimes within that window.

C.3. Model hyperparameters

The TFT model architecture is configured with the following hyperparameters, optimized through systematic random search on the validation set. *Architecture Parameters*: Hidden layer size of 256 units, 4 attention heads, dropout rate of 0.3 applied to all layers, single LSTM layer for encoder and decoder, estimation horizon (encoder length) of 72 months, and prediction horizon (decoder length) of 12 months. *Training*

Parameters: Batch size of 128, learning rate of 0.001 with Adam optimizer, ReduceLROnPlateau scheduler (factor=0.5, patience=5), gradient clipping at 0.1, early stopping patience of 10 epochs, maximum of 100 epochs, and quantile loss function predicting 10th, 50th, and 90th percentiles. *Data Requirements*: Minimum sequence length of 84 months (72 encoder + 12 decoder), maximum encoder length of 72 months, and monthly cross-sectional standardization (zero mean, unit variance).

C.4. Hyperparameter optimization procedure

For each candidate hyperparameter configuration in the search space, the optimization procedure trains the model on all k=10 folds using the candidate configuration, evaluates performance on each fold's 20% validation subset using quantile loss, computes average validation loss across all 10 folds, and selects the configuration with lowest average validation loss. The search space explores hidden size {128, 256, 512}, attention heads {2, 4, 8}, dropout {0.1, 0.2, 0.3, 0.4}, learning rate {0.0001, 0.001, 0.01}, and batch size {64, 128, 256}.

C.5. Ensemble prediction methodology

Final predictions are generated through an ensemble approach leveraging the overlapping test periods across folds. For each fold, the model is trained on its full training period (all funds, no validation split) using optimal hyperparameters and generates predictions for each fund-month in the fold's test period. For each fund-month observation appearing in multiple folds' test periods, predictions from all applicable folds are averaged. For example, a fund-month observation in January 2015 receives predictions from Folds 1–6 (all folds whose test periods include 2015), and the final prediction is the average across these 6 forecasts. This ensemble averaging substantially reduces prediction variance and improves stability.

C.6. Data preprocessing and feature engineering

All input variables undergo standardization before entering the model. Dynamic variables are standardized monthly using cross-sectional mean and standard deviation (calculated separately for each calendar month across all funds). Static variables are encoded as categorical variables (fund style, load indicator). Deterministic variables are normalized to [0,1] range (month of year, manager tenure). Standardization parameters (means, standard deviations) are calculated exclusively from each fold's training data and applied consistently to validation and test data, preventing information leakage.

C.7. Computational resources

Model training was conducted on high-performance computing infrastructure: GPU — NVIDIA L40S (46 GB VRAM); System Memory — 128 GB RAM; CPU — 64 cores for parallel data loading; Storage — SSD with 100 GB available space; Training time per fold — Approximately 45–60 min; Total training time (11 folds, sequential) — Approximately 10 h.

C.8. Software implementation

The model is implemented using the following software stack: Python 3.9, PyTorch 2.5.1 (with CUDA 12.1 support), PyTorch Lightning 2.5.6, PyTorch Forecasting 1.4.0, data processing via pandas 1.5.3 and numpy 1.24.3, and HDF5 storage using h5py 3.8.0 and tables 3.8.0. All code is available upon request to facilitate replication.

Appendix D. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jfs.2026.101532>.

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